**ROLL NO:** _____

# MADHAV INSTITUTE OF TECHNOLOGY & SCIENCE, GWALIOR

(A Deemed to be University | Autonomous Institute Established in 1957 | NAAC 'A++' Accredited)

## END-SEM EXAMINATION — DECEMBER / JUNE 2024

FIRST YEAR (COMMON TO ALL BRANCHES) — SEMESTER 2

**Time Allowed: 3 Hours** **Maximum Marks: 70**

*NOTE: 1. Attempt any FIVE questions selecting at least ONE question from each Unit.*
*2. All questions carry equal marks. Sub-questions carry marks indicated in brackets [ ].*
*3. Assume suitable missing data, if any, and state it clearly.*
*4. Use of scientific non-programmable calculator is permitted.*

---

**--- UNIT I ---**

**Q.1** Describe Building Materials: Composition and manufacturing of Portland Cement. Field tests on bricks. [14 Marks]

**[ OR ]**

**Q.1 (Alt)** Explain Concrete grades, Workability (Slump test), Water-Cement ratio, and Compressive strength testing. [14 Marks]

**--- UNIT II ---**

**Q.2** Explain Principles of Surveying: Chain Surveying, Compass Surveying, Whole Circle Bearing vs Quadrantal Bearing. [14 Marks]

**[ OR ]**

**Q.2 (Alt)** Explain Leveling: Height of Instrument method vs Rise and Fall method. Calculate Reduced Levels (RL). [14 Marks]

**--- UNIT III ---**

**Q.3** State and prove Lami's Theorem and Varignon's Theorem of Moments. Find resultant of concurrent forces. [14 Marks]

**[ OR ]**

**Q.3 (Alt)** Calculate Centroid and Moment of Inertia for standard T-section and I-section. [14 Marks]

**--- UNIT IV ---**

**Q.4** State Laws of Dry Friction (Coulomb's Friction). Solve problems involving blocks on rough inclined planes. [14 Marks]

**[ OR ]**

**Q.4 (Alt)** Analyze simple pin-jointed trusses using Method of Joints and Method of Sections. [14 Marks]

**--- UNIT V ---**

**Q.5** Explain Building Components: Substructure (Shallow vs Deep foundations) and Superstructure (Walls, Columns, Beams). [14 Marks]

**[ OR ]**

**Q.5 (Alt)** Describe Green Buildings, Rainwater Harvesting techniques, and basic site planning principles. [14 Marks]

**** END OF QUESTION PAPER — MITS GWALIOR AUTONOMOUS EXAMINATION CELL ****